

BIG DATA PIPELINE

Using Hadoop Ecosystem

Olist E-Commerce Dataset • End-to-End ETL Project

Hadoop

Apache Sqoop

HDFS

Apache Hive

MapReduce

PostgreSQL



Project Report



Agenda

01

Project Overview

Goals, dataset, and scope

02

Pipeline Architecture

End-to-end flow & components

03

Step-by-Step Implementation

PostgreSQL → HDFS → Hive → MapReduce

04

Analytics & Hive Queries

Business-level SQL analytics

05

Key Results & Insights

Demo outputs & findings

06

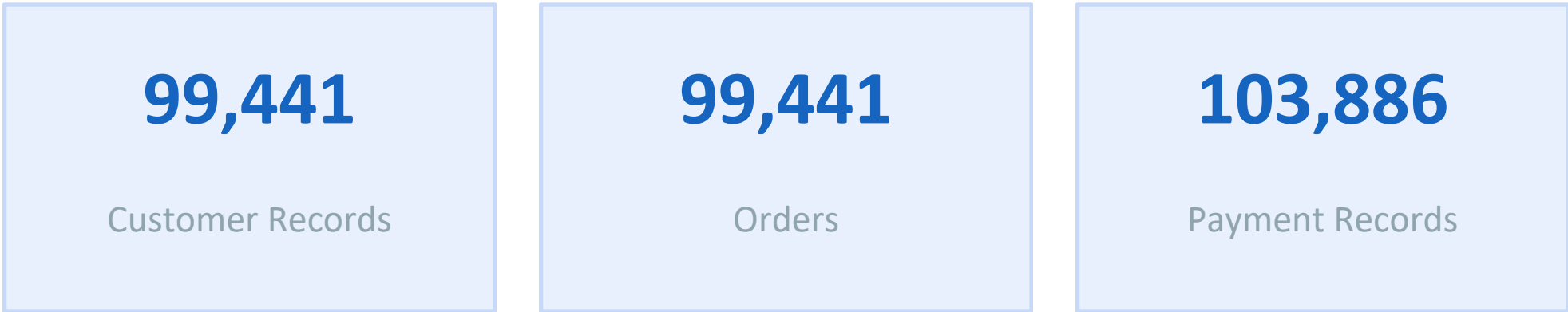
Future Improvements

Roadmap & next steps

Project Overview

What is this project?

An end-to-end Big Data ETL pipeline that ingests, cleans, stores, and analyzes real-world Brazilian e-commerce data using the Hadoop ecosystem.



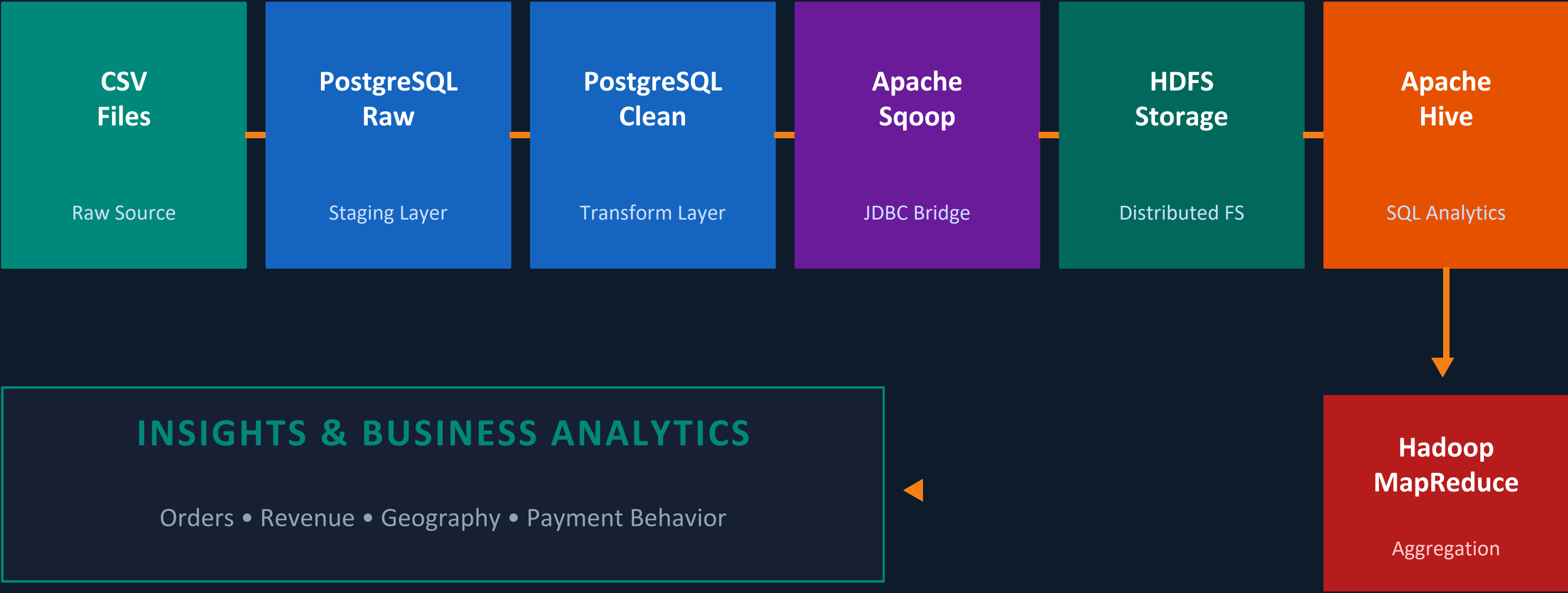
Project Goals

- Ingest CSV data via PostgreSQL staging tables
- Transfer cleaned data to HDFS via Sqoop
- Run HiveQL analytics on distributed data
- Process data with Hadoop Streaming MapReduce

Dataset Files

File	Rows	Use
customers_dataset	99,441	Geography
orders_dataset	99,441	Lifecycle
order_payments	103,886	Revenue

Pipeline Architecture



SOURCE

STAGING

TRANSPORT

PROCESSING

Data Transformation — Raw → Clean

RAW TABLES

customers_raw

customer_id
customer_unique_id
zip_code_prefix ✗
customer_city
customer_state

orders_raw

order_id
customer_id
order_status
purchase_timestamp
+ 4 more timestamp cols ✗

payments_raw

order_id
payment_sequential ✗
payment_type
payment_installments ✗
payment_value

SELECT
only needed
cols



CLEAN TABLES

customers_clean

✓ customer_id
✓ customer_city
✓ customer_state

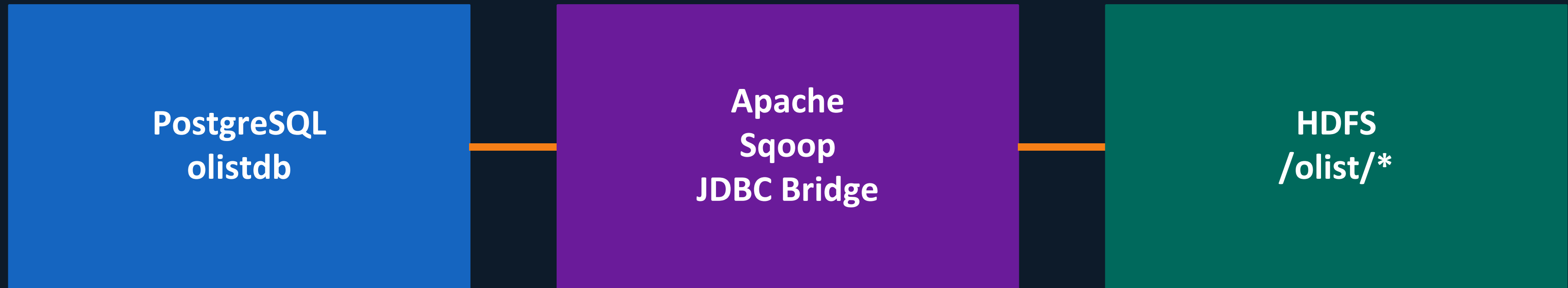
orders_clean

✓ order_id
✓ customer_id
✓ order_status
✓ order_purchase_timestamp

payments_clean

✓ order_id
✓ payment_type
✓ payment_value

Sqoop Import → HDFS



HDFS Directory Structure

`/olist/`

```
|— customers/    → part-m-00000  (99,441 rows)
|— orders/       → part-m-00000  (99,441 rows)
|— payments/     → part-m-00000  (103,886 rows)
└— output_payment_count/ → part-00000  (MapReduce result)
```

```
sqoop import --connect jdbc:postgresql://localhost/olistdb --table customers_clean --target-dir /olist/customers -m 1
```

Apache Hive — Analytics Queries

Total Orders

```
SELECT COUNT(*) FROM orders;
```

Orders by Status

```
SELECT order_status, COUNT(*)  
FROM orders GROUP BY order_status;
```

Top States by Customers

```
SELECT customer_state, COUNT(*)  
FROM customers GROUP BY customer_state  
ORDER BY COUNT(*) DESC LIMIT 10;
```

Revenue by Payment Type

```
SELECT payment_type, SUM(payment_value)  
FROM payments GROUP BY payment_type;
```

★ Multi-table JOIN: orders ⋈ customers ⋈ payments → State-wise Revenue Analysis

Hadoop Streaming MapReduce

HDFS Input
/olist/payments

mapper.py
emit payment_type → 1

Shuffle &
Sort (Hadoop)

reducer.py
sum counts

mapper.py

```
for line in sys.stdin:
    data = line.strip().split(',')
    if len(data) > 2:
        payment_type = data[1]
        print(f'{payment_type}\t1')
```

reducer.py

```
for line in sys.stdin:
    payment_type, count = line.split('\t')
    if current_type == payment_type:
        current_count += int(count)
    else: print(current_type, current_count)
```

Output:

```
credit_card 76795   boleto 19784   voucher 5775   debit_card 1529
```


Key Results & Insights

99,441

Total Orders

97% delivered

0.6%

Cancellation Rate

Excellent fulfillment

73.9%

Credit Card Revenue

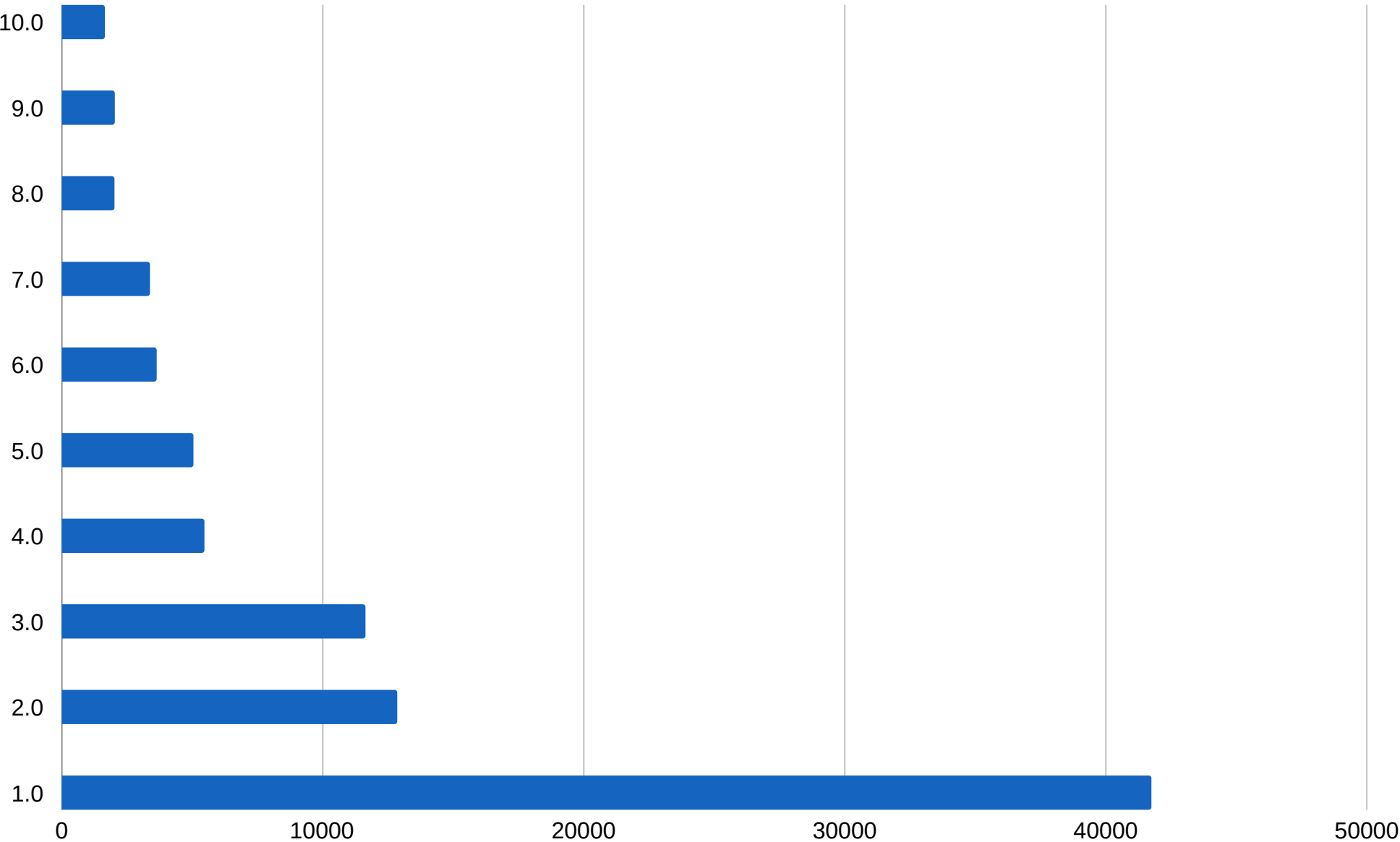
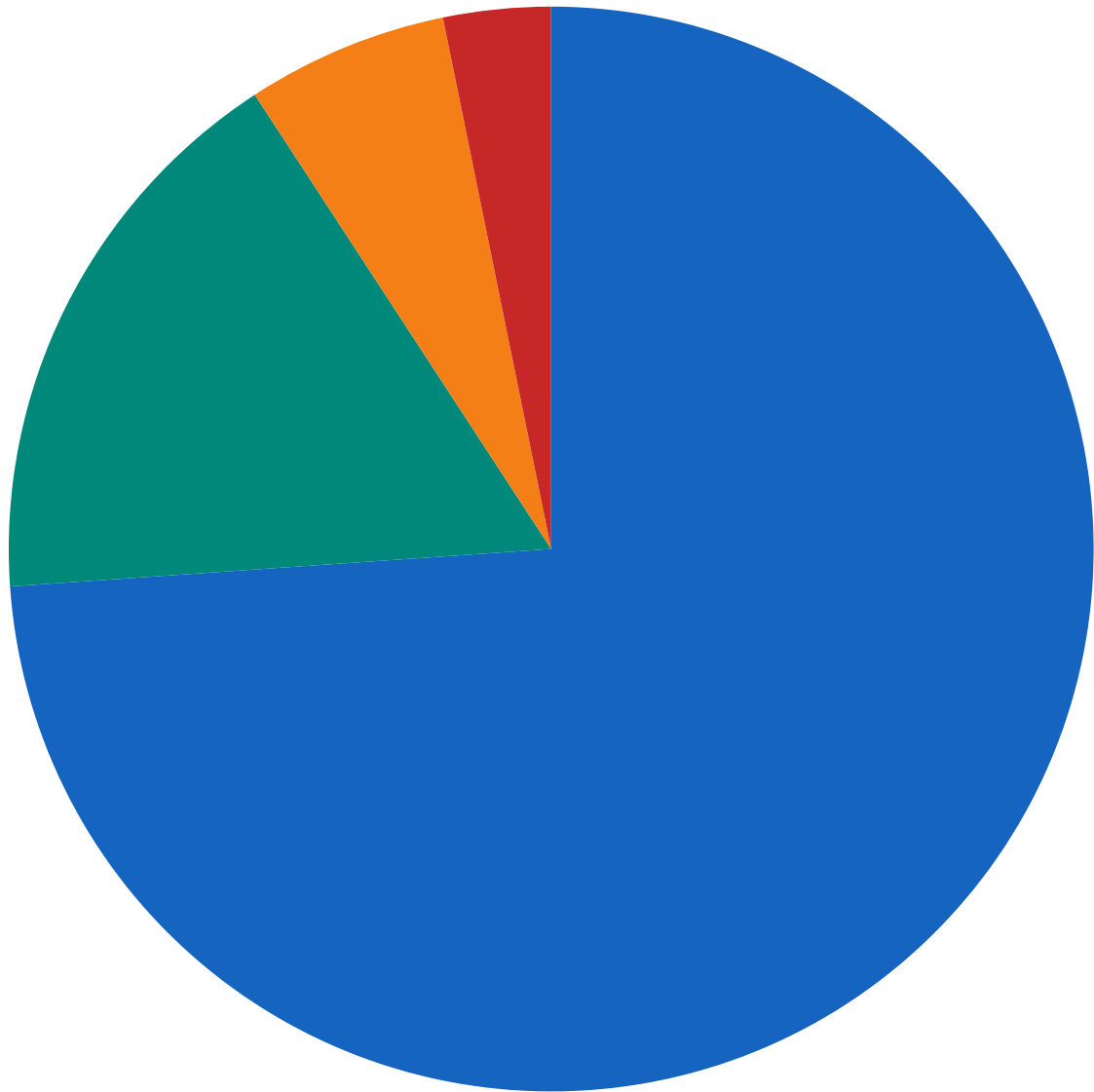
Dominant payment mode

42%

Customers from SP

São Paulo dominates

Credit Card Boleto Voucher Debit Card



Business Insights Summary

01	Geography São Paulo accounts for 42% of customers & ~37% of total revenue	High Concentration
02	Payment Credit card = 74% of revenue; Boleto = 17% as an alternative	Credit Dominant
03	Fulfillment 97% of orders successfully delivered — strong logistics	Excellent
04	Spend Parity Avg spend R\$136–R\$150 consistent across all states — no regional pricing gap	Uniform Pricing
05	Cancellations Only 0.6% cancellation rate — good inventory & payment processing	Low Risk
06	Validation MapReduce output independently confirms Hive analytics results	Pipeline Reliable

Thank You

Hadoop Big Data Pipeline — Olist Dataset

- ✓ PostgreSQL staging & data cleaning
- ✓ HDFS distributed storage via Sqoop
- ✓ Hive SQL analytics on distributed data
- ✓ Hadoop Streaming MapReduce in Python
- ✓ Business insights from real-world e-commerce data

End-to-End ETL • Real-World Big Data Concepts